Bottle Washing Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bottle washing machine

What it is used for

A bottle washing machine cleans reusable bottles by using water, approved detergents, rinsing, and mechanical or spray action. Operate it according to the bottle type and sanitation procedure to remove residues and reduce contamination before filling.

Main parts

- Machine frame and bottle holders or conveyor
- Wash and rinse tanks or spray nozzles
- Water supply and drain connections
- Detergent dosing system, if fitted
- Pump, motor, and drive mechanism
- Control panel, guards, and emergency stop

Before you start

1. Confirm the bottle type, washing capacity, and required cleaning or sanitation procedure.
2. Inspect the conveyor, bottle holders, spray nozzles, hoses, tanks, guards, and electrical connections.
3. Remove broken glass, loose debris, and unsuitable bottles from the work area.
4. Check that water quality, detergent, and any sanitising agent are approved for the intended food-contact application.
5. Verify that wash and rinse tanks contain the required water and that drains are clear.
6. Wear suitable gloves, safety footwear, and eye protection when handling chemicals or glass bottles.

How to use

1. Switch off and isolate the machine before opening guards, clearing jams, or cleaning internal components.
2. Remove remaining liquid and loose debris from bottles using the approved pre-rinse procedure, where required.
3. Load bottles into the designated holders or conveyor with stable spacing and correct orientation.
4. Prepare detergent or cleaning solution at the concentration specified by the manufacturer and site procedure.
5. Select the correct wash cycle, temperature, pressure, and conveyor speed for the bottle type.
6. Close guards and start the machine using the approved controls.
7. Observe bottle movement, spray coverage, water flow, and machine operation during the cycle.

8. Check that bottles receive the required wash and rinse stages without damage or tipping.
9. Stop the machine if bottles jam, glass breaks, leakage occurs, or abnormal noise develops.
10. Allow the cycle to finish and unload bottles carefully into the designated clean area.
11. Inspect a representative sample for visible residue and verify required sanitation checks according to the facility procedure.
12. Drain and clean the machine as specified, then record cleaning results and any faults.

⚠ Safety

- Only trained operators should operate or maintain the machine.
- Never reach into moving conveyors, bottle holders, or spray mechanisms.
- Isolate power and any relevant water or chemical supply before clearing jams or servicing equipment.
- Use appropriate PPE when handling cleaning chemicals, hot water, or broken glass.
- Do not mix cleaning chemicals unless the approved procedure explicitly permits it.
- Keep electrical components protected from water and report damaged cables or guards immediately.

Care & storage

- Clean wash tanks, filters, spray nozzles, and bottle holders according to the sanitation schedule.
- Inspect hoses, seals, pumps, conveyor components, and chemical dosing equipment regularly.
- Remove mineral deposits and detergent residue using approved cleaning methods.
- Check temperature controls and dosing accuracy at specified intervals.
- Arrange servicing for leaks, poor cleaning performance, or repeated conveyor faults.

Common mistakes to avoid

- Loading bottles incorrectly or overloading the conveyor.
- Using the wrong detergent concentration or wash settings.
- Ignoring blocked spray nozzles or dirty rinse water.
- Handling broken glass without proper protection or reaching into a moving machine.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.